

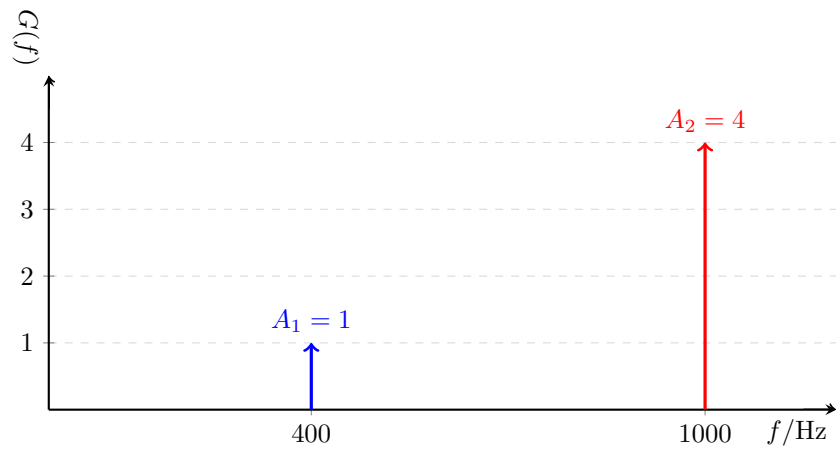
CSCI2750 Homework 1

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1 Problem 1

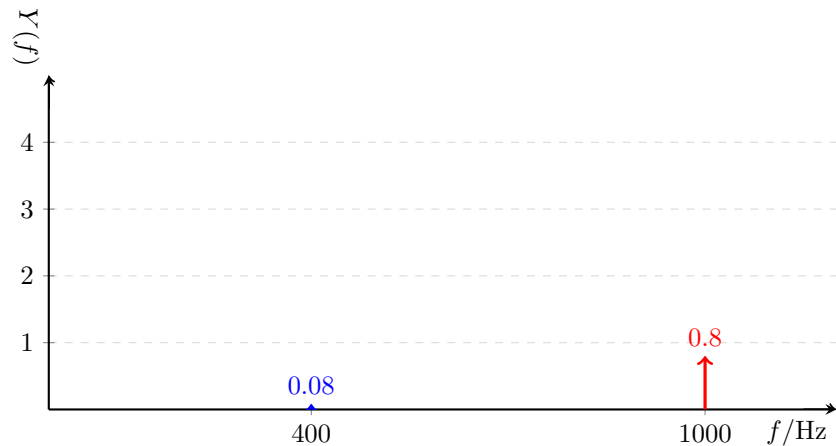
1.1 Part (a)



1.2 Part (b)

$$H(f_1) = \frac{400}{5000} = 0.08$$
$$H(f_2) = \frac{1000}{5000} = 0.2$$

1.3 Part (c)

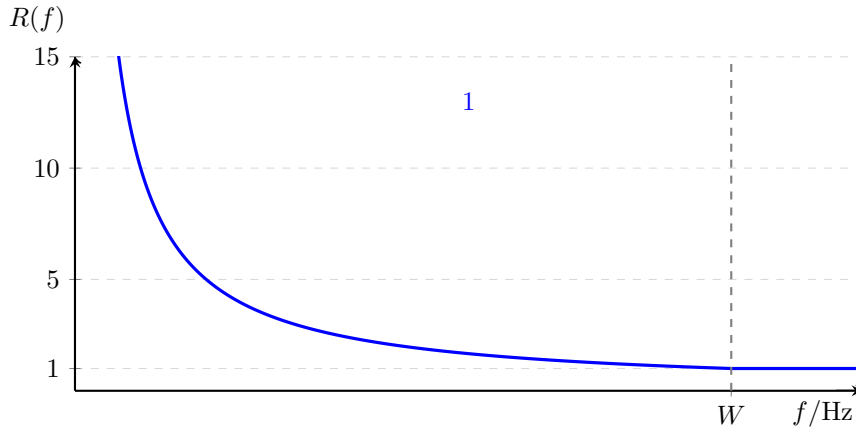


1.4 Part (d)

for equalizer $R(f)$, we need to find a function that will make the output signal $Y(f) = X(f) * H(f)$.

$$R(f) = \frac{1}{H(f)}$$

The plot of $R(f)$ is as follows:

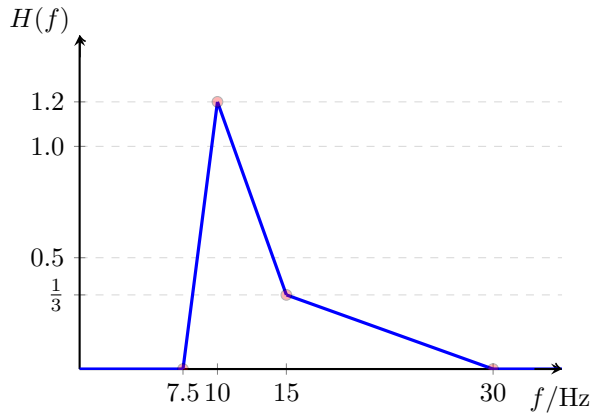


For 400 Hz, amplification needed = $\frac{1}{0.08} = 12.5$
 For 1000 Hz, amplification needed = $\frac{1}{0.2} = 5$

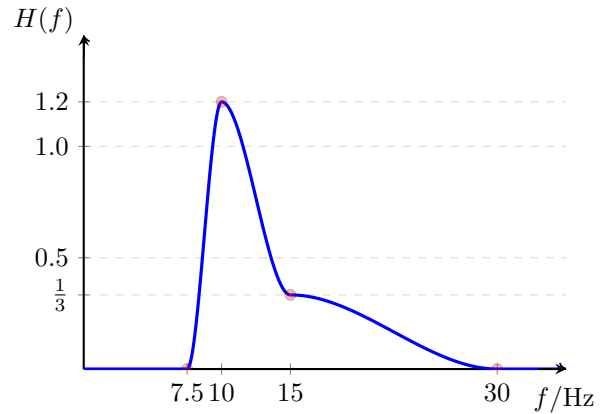
2 Problem 2

2.1 Part (a)

Two possible filter responses that satisfy the input-output relationship:



(a) Filter Response 1



(b) Filter Response 2

2.2 Part (b)

$$v(t) = w(t) \cos(120\pi t) = [2 \sin(40\pi t) + \sin(20\pi t)] \cos(120\pi t)$$

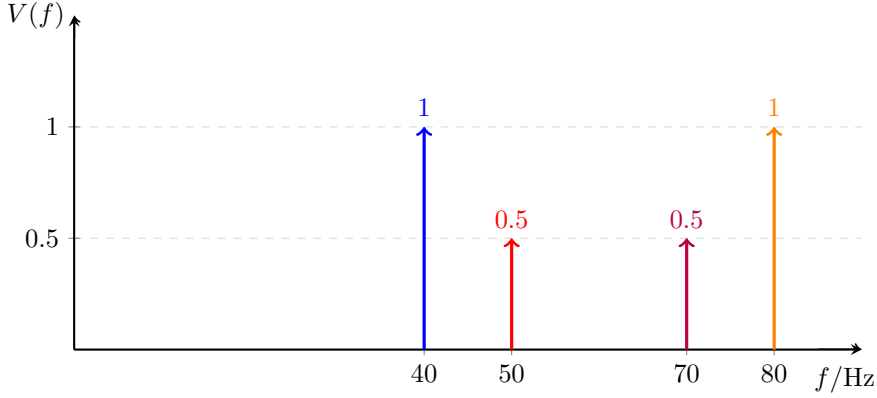
$$v(t) = 2 \sin(40\pi t) \cos(120\pi t) + \sin(20\pi t) \cos(120\pi t)$$

$$v(t) = \sin(160\pi t) - \sin(80\pi t) + \frac{1}{2} \sin(140\pi t) - \frac{1}{2} \sin(100\pi t)$$

The frequency components of $v(t)$ are:

- $f = 40$ Hz: amplitude $|1| = 1$
- $f = 50$ Hz: amplitude $|\frac{1}{2}| = 0.5$
- $f = 70$ Hz: amplitude $|\frac{1}{2}| = 0.5$
- $f = 80$ Hz: amplitude $|1| = 1$

The spectrum of $v(t)$ is as follows:



3 Problem 3

3.1 Part (a)

In this case, f_o is the sampling frequency of $g(t)$.

3.2 Part (b)

Given $x(t) = \sum_{j=0}^J B_j \sin(2\pi j f_x t)$. The bandwidth of $x(t)$ is:

$$W = J f_x$$

According to Nyquist sampling theorem,

$$f_0 \geq 2W = 2J f_x$$

Therefore, a proper value for f_0 is:

$$f_0 = 2J f_x$$

3.3 Part (c)

Given $J = 3$, $B_j = 1/(j+1)^2$, $A_k = 1$ for all k .

Calculating the amplitudes of $x(t)$:

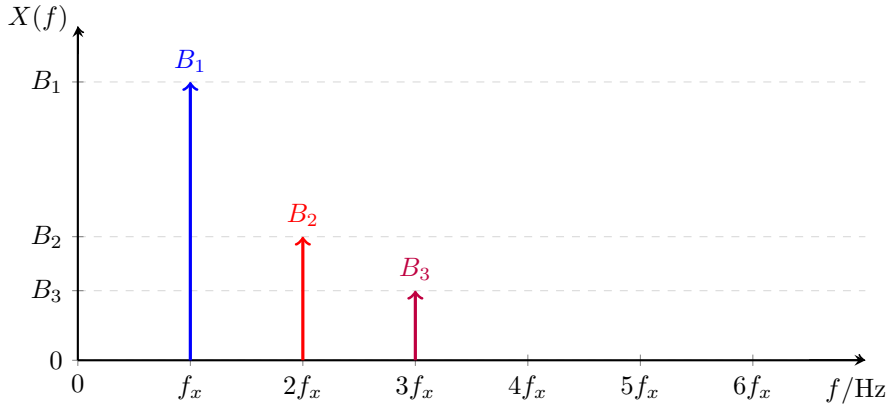
$$B_0 = \frac{1}{(0+1)^2} = 1 \quad B_1 = \frac{1}{(1+1)^2} = \frac{1}{4} \quad B_2 = \frac{1}{(2+1)^2} = \frac{1}{9} \quad B_3 = \frac{1}{(3+1)^2} = \frac{1}{16}$$

Therefore, $x(t)$ has frequency components at:

- $f = 0$: $\sin(0) = 0$
- $f = f_x$: amplitude $B_1 = 1/4$
- $f = 2f_x$: amplitude $B_2 = 1/9$
- $f = 3f_x$: amplitude $B_3 = 1/16$

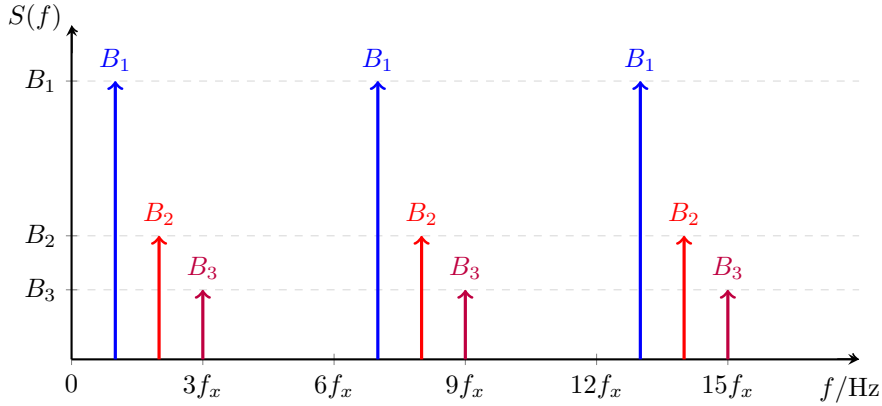
bandwidth $W = 3f_x$. From Part (b), $f_0 = 2W = 6f_x$.

Frequency-domain spectrum of $x(t)$ (one-sided):

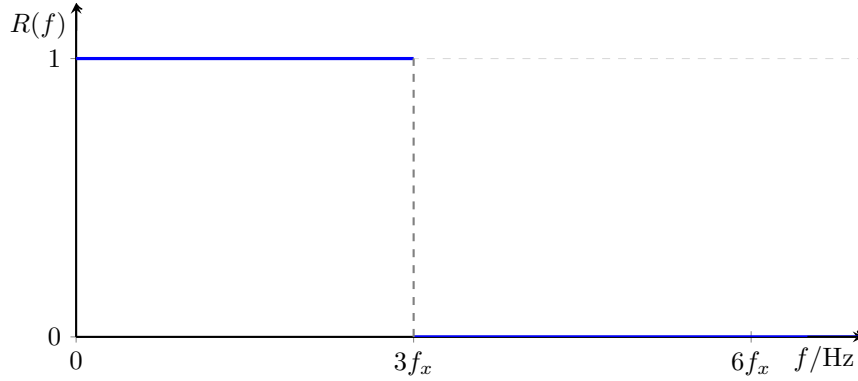


Frequency-domain spectrum of $s(t) = x(t) \cdot g(t)$ (one-sided):

(Note: The spectrum of $s(t)$ is a graph of $x(t)$ repeated every $6f_x$, i.e. f_0 .)



Ideal low-pass filter $R(f)$ to recover $x(t)$ from $s(t)$:



The filter passes frequencies from 0 to $3f_x$ (the base bandwidth) and rejects all replicas at higher frequencies.

4 Problem 4

4.1 Part (a)

For $m(t)$, bandwidth $W = 50\text{Hz}$. Thus, the Nyquist rate $f_s = 2W = 100\text{Hz}$.

4.2 Part (b)

Dynamic range $= \frac{A_{max}}{A_{min}} = \frac{1.7}{0.3} = 5.67$
 $m(t)$ spanning $[-3, 3]$.

4.3 Part (c)

Since range of sin function $= [-1, 1]$, the range of $m(t) = [-3, 3]$.

We can see when $m(t) = -2$ or 0 or 2 , the error is the largest because they are in the middle of the quantization levels.

i.e. largest quantization error $= 0.5$

4.4 Part (d)

Assuming $f_o = 100\text{Hz}$, sample period $= 0.01\text{s}$.

For $t = 1.02$, $m(t) \approx 0.898$ For $t = 1.03$, $m(t) \approx -0.765$ For $t = 1.04$, $m(t) \approx 0.575$ For $t = 1.05$, $m(t) \approx 2.569$

Quantized value $= [1, -1, 1, 3]$

4.5 Part (e)

4-level quantizer: 2 bit per sample, 100 samples per second.

Total bits needed $= 2 * 100 * 45 = 9000\text{bits}$

5 Problem 5

5.1 Part (a)

Value 0: 17 times, Value 50: 8 times.

Histogram:



5.2 Part (aii)

$$G = \frac{1}{11} \begin{bmatrix} 50 & 150 & 150 & 100 & 0 \\ 50 & 300 & 350 & 400 & 100 \\ 100 & 400 & 350 & 350 & 150 \\ 50 & 200 & 400 & 300 & 150 \\ 50 & 50 & 100 & 50 & 50 \end{bmatrix} \approx \begin{bmatrix} 4.545 & 13.64 & 13.64 & 9.091 & 0 \\ 4.545 & 27.27 & 31.82 & 36.36 & 9.091 \\ 9.091 & 36.36 & 31.82 & 31.82 & 13.64 \\ 4.545 & 18.18 & 36.36 & 27.27 & 13.64 \\ 4.545 & 4.545 & 9.091 & 4.55 & 4.545 \end{bmatrix}$$

5.3 Part (bi)

By observation, most values in I is between 90 and 100. However, in $I[2,2]$ and $I[4,4]$ the values are either larger or smaller than this range for a huge difference.

Therefore, if we perform median filtering, the values of $I[2,2]$ and $I[4,4]$ will be replaced by the median of the surrounding values. the sharp edge will be blurred.

5.4 Part (bii)

$$J = \begin{bmatrix} 0 & 98 & 98 & 97 & 0 \\ 98 & 98 & 98 & 98 & 96 \\ 96 & 98 & 98 & 97 & 94 \\ 95 & 97 & 96 & 97 & 50 \\ 0 & 95 & 50 & 50 & 0 \end{bmatrix}$$

5.5 Part (biii)

for $n \times n$ median filtering with zero padding, the data in the corner will be replaced by zero. Also, border pixels will be biased towards zero.